UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING – Written Exam 28/01/2025

PROBLEM - We observe two samples of the discrete-time real-valued w.s.s. random process $X[n]=S[n]+W[n]$ at $n=0$ and $n=2$, where $S[n]$ is a w.s.s. zero-mean Gaussian process modelled as an Auto-Regressive (AR) process of order 1, i.e. $S[n]=\rho S[n-1]+I[n]$, where ρ is the one-lag correlation coefficient of the process $S[n]$ and $I[n]$ is the white innovation process with power σ_I^2. $W[n]$ models Additive White Gaussian Noise (AWGN) with Power Spectral Density (PSD) $S_W\left(e^{j2\pi f}\right)=\sigma_W^2$; $W[n]$ and $S[n]$ are mutually independent.

(1) Derive the expressions of the power σ_X^2, the autocorrelation function (ACF) $R_X[m]$, and the Power Spectral Density (PSD) $S_X\left(e^{j2\pi f}\right)$ of the process $X[n]$. Derive the Signal to Noise Ratio (SNR) $\gamma=E\left\{S^2[n]\right\}\big/E\left\{W^2[n]\right\}$. Calculate all these quantities for $\sigma_I^2=10$, $\sigma_W^2=2$, and $\rho=-0.9$, plot the ACF and the PSD.

(2) Derive the covariance matrix of the 2-dimensional data vector $\mathbf{X}=[X[0]\,X[2]]^T$ and its probability density function (PDF).

(3) Solve the following **interpolation in noise** problem: Assuming that for some reason we could not observe $S[1]$, find the optimal (in the mean square sense) linear interpolator that retrieves (estimates) $S[1]$ based on the observation of the two adjacent noisy samples $X[0]$ and $X[2]$.

(4) Derive the bias $bias=E\left\{S[1]-\hat{S}[1]\right\}$ and the *Mean Square Error* (MSE) $\sigma_\varepsilon^2=E\left\{\left(S[1]-\hat{S}[1]\right)^2\right\}$

$X[n]=S[n]+W[n]$ of the optimal linear estimator $\hat{S}[1]$ as a function of γ and ρ; plot the MSE versus γ, analyse what happens to $\hat{S}[1]$ and σ_ε^2 when $\gamma\to 0$ and $\gamma\to+\infty$ and comment the results.

Solution

(1) <u>Power, ACF, PSD and SNR.</u>

$$\sigma_X^2 = E\left\{X^2[n]\right\} = E\left\{S^2[n]\right\} + E\left\{W^2[n]\right\} + E\left\{S[n]W[n]\right\} = \sigma_S^2 + \sigma_W^2$$

$$R_X[m] = E\left\{X[n]X[n+m]\right\} = R_S[m] + R_W[m] = R_S[m] + \sigma_W^2\delta[m]$$

$$S_X\left(e^{j2\pi f}\right) = S_S\left(e^{j2\pi f}\right) + S_W\left(e^{j2\pi f}\right) = S_S\left(e^{j2\pi f}\right) + \sigma_W^2$$

$$S[n] = \rho S[n-1] + I[n]$$

$$S(z) = \rho z^{-1}S(z) + I(z) \quad \Rightarrow \quad L(z) = \frac{S(z)}{I(z)} = \frac{1}{1-\rho z^{-1}} = \frac{z}{z-\rho}$$

The innovation filter has 1 zero at $z=0$ and 1 pole at $z=\rho$. Hence, the innovation filter is a high-pass filter if $\rho<0$ and a low-pass filter if $\rho>0$, and so the process. The ACF and the PSD are given by (see the course vu-graphs for the derivation):

$$S_S\left(e^{j2\pi f}\right) = \sigma_I^2\left|L\left(e^{j2\pi f}\right)\right|^2 = \sigma_I^2\left|\frac{1}{1-\rho e^{-j2\pi f}}\right|^2 = \frac{\sigma_I^2}{1+\rho^2-2\rho\cos(2\pi f)}$$

$$R_S[m] = FT^{-1}\left\{S_S\left(e^{j2\pi f}\right)\right\} = \frac{\sigma_I^2}{1-\rho^2}\rho^{|m|}$$

Hence $\sigma_S^2 = R_S[0] = \dfrac{\sigma_I^2}{1-\rho^2} \;\rightarrow\; \sigma_X^2 = \sigma_S^2 + \sigma_W^2 = \dfrac{\sigma_I^2}{1-\rho^2} + \sigma_W^2$

$$R_X[m] = \frac{\sigma_I^2}{1-\rho^2}\rho^{|m|} + \sigma_W^2\delta[m], \quad S_X\left(e^{j2\pi f}\right) = \frac{\sigma_I^2}{1+\rho^2-2\rho\cos(2\pi f)} + \sigma_W^2$$

As previously stated, when $\rho<0$, $S[n]$ is a high-pass process, when $\rho>0$, $S[n]$ is a low-pass process.

$$\gamma = \frac{E\left\{S^2[n]\right\}}{E\left\{W^2[n]\right\}} = \frac{\sigma_I^2}{\left(1-\rho^2\right)\sigma_W^2}$$

(2) <u>Covariance matrix of the data vector</u> $\mathbf{X} = [X[0]\; X[2]]^T$.

The covariance matrix $\mathbf{C}_X = E\left\{\mathbf{X}\mathbf{X}^T\right\}$ is as follows:

$$\mathbf{C}_X = E\left\{\mathbf{X}\mathbf{X}^T\right\} = E\left\{(\mathbf{S}+\mathbf{W})(\mathbf{S}+\mathbf{W})^T\right\} = \mathbf{C}_S + \mathbf{C}_W = \mathbf{C}_S + \sigma_W^2\mathbf{I}$$

$$\mathbf{C}_S = E\left\{\mathbf{S}\mathbf{S}^T\right\} = E\left\{\begin{bmatrix} S[0] \\ S[2] \end{bmatrix}[S[0]\; S[2]]\right\} = E\left\{\begin{bmatrix} S^2[0] & S[0]S[2] \\ S[2]S[0] & S^2[2] \end{bmatrix}\right\} = \begin{bmatrix} R_S[0] & R_S[2] \\ R_S[-2] & R_S[0] \end{bmatrix} = \sigma_S^2\begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix}$$

$$\mathbf{C}_X = \mathbf{C}_S + \sigma_W^2\mathbf{I} = \sigma_S^2\begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix} + \sigma_W^2\mathbf{I} = \begin{bmatrix} \sigma_S^2+\sigma_W^2 & \sigma_S^2\rho^2 \\ \sigma_S^2\rho^2 & \sigma_S^2+\sigma_W^2 \end{bmatrix} = \sigma_W^2\begin{bmatrix} 1+\gamma & \gamma\rho^2 \\ \gamma\rho^2 & 1+\gamma \end{bmatrix}$$

The pdf of $\mathbf{X}$ can be derived by observing that $X[n]$ is a Gaussian process, so $\mathbf{X}$ is a Gaussian vector:

$\mathbf{X} \in N\left(\boldsymbol{\eta}_X, \mathbf{C}_X\right)$, where $\boldsymbol{\eta}_X = E\{\mathbf{X}\} = \mathbf{0}$ and $\mathbf{C}_X = \sigma_W^2\begin{bmatrix} 1+\gamma & \gamma\rho^2 \\ \gamma\rho^2 & 1+\gamma \end{bmatrix}$.

(3) LMMSE estimator

Since the process is Gaussian, the LMMSE coincides with the MMSE and the MAP estimators.
The LMMSE estimator is obtained by imposing that the estimator is unbiased and applying the Orthogonality Principle.

$\hat{S}[1] = a_1X[0] + a_2X[2] + b, \quad \varepsilon[1] = S[1] - \hat{S}[1]$

Since $S[n]$ and $X[n]$ are zero mean processes, imposing that the estimator is unbiased, we get:

$E\{\varepsilon[1]\} = E\left\{S[1] - \hat{S}[1]\right\} = 0 \quad \Rightarrow \quad b = 0$.

Orthogonality principle:

$E\{\varepsilon[1]X[1+m]\} = 0, \quad m = -1,1$

$E\left\{\left(S[1] - \hat{S}[1]\right)X[1+m]\right\} = 0, \quad m = -1,1$

$E\left\{\left(S[1] - a_1X[0] - a_2X[2]\right)X[1+m]\right\} = 0, \quad m = -1,1$

$R_S[m] - a_1R_X[m+1] - a_2R_X[m-1] = 0, \quad m = -1,1$

$$\begin{cases} a_1R_X[0] + a_2R_X[-2] = R_S[-1] \\ a_1R_X[2] + a_2R_X[0] = R_S[1] \end{cases}$$

$$\begin{bmatrix} R_X[0] & R_X[-2] \\ R_X[2] & R_X[0] \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} R_S[-1] \\ R_S[1] \end{bmatrix}$$

$$R_X[0] = \sigma_S^2 + \sigma_W^2 = \sigma_W^2\left(1+\gamma\right), \quad R_S[1] = R_S[-1] = \sigma_S^2\rho, \quad R_X[-2] = R_X[2] = \sigma_S^2\rho^2$$

$$\sigma_W^2 \begin{bmatrix} 1+\gamma & \gamma\rho^2 \\ \gamma\rho^2 & 1+\gamma \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \sigma_S^2\begin{bmatrix} \rho \\ \rho \end{bmatrix} \quad \rightarrow \quad \begin{bmatrix} 1+\gamma & \gamma\rho^2 \\ \gamma\rho^2 & 1+\gamma \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \gamma\rho\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \gamma\rho\begin{bmatrix} 1+\gamma & \gamma\rho^2 \\ \gamma\rho^2 & 1+\gamma \end{bmatrix}^{-1}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{\gamma\rho}{(1+\gamma)^2 - \gamma^2\rho^4}\begin{bmatrix} 1+\gamma & -\gamma\rho^2 \\ -\gamma\rho^2 & 1+\gamma \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{\gamma\rho\left(1+\gamma-\gamma\rho^2\right)}{(1+\gamma)^2 - \gamma^2\rho^4}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{\gamma\rho}{1+\gamma+\gamma\rho^2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{S}[1] = \frac{\gamma\rho}{1+\gamma+\gamma\rho^2}X[0] + \frac{\gamma\rho}{1+\gamma+\gamma\rho^2}X[2] = \frac{2\gamma\rho}{1+\gamma+\gamma\rho^2}\left(\frac{X[0]+X[2]}{2}\right) = \alpha\cdot\frac{X[0]+X[2]}{2}$$

where $\alpha = \dfrac{2\gamma\rho}{1+\gamma+\gamma\rho^2}$.

(4) <u>Bias and MSE</u>

$$E\left\{\varepsilon[1]\right\} = E\left\{S[1] - \hat{S}[1]\right\} = 0 \quad \Rightarrow \quad bias = 0$$

$$\begin{aligned}\sigma_\varepsilon^2 &= E\left\{\varepsilon^2[1]\right\} = E\left\{\left(S[1]-\hat{S}[1]\right)S[1]\right\} = E\left\{\left(S[1] - \frac{\alpha}{2}X[0] - \frac{\alpha}{2}X[2]\right)S[1]\right\} \\ &= R_S[0] - \frac{\alpha}{2}R_S[1] - \frac{\alpha}{2}R_S[-1] = \sigma_S^2 - \alpha\sigma_S^2\rho \\ &= \sigma_S^2\left(1-\alpha\rho\right) = \sigma_S^2\left(1 - \frac{2\gamma\rho^2}{1+\gamma+\gamma\rho^2}\right) = \sigma_S^2\frac{1+\gamma-\gamma\rho^2}{1+\gamma+\gamma\rho^2}\end{aligned}$$

In summary: $\hat{S}[1] = \dfrac{2\gamma\rho}{1+\gamma+\gamma\rho^2}\left(\dfrac{X[0]+X[2]}{2}\right)$, $\sigma_\varepsilon^2 = \sigma_S^2\dfrac{1+\gamma-\gamma\rho^2}{1+\gamma+\gamma\rho^2}$

$\gamma \rightarrow 0$: $\hat{S}[1] \rightarrow 0$, which is the a-priori mean of $S[1]$, and $\sigma_\varepsilon^2 \rightarrow \sigma_S^2$ which is the a-priori variance of $S[1]$.

$\gamma \rightarrow +\infty$: $\hat{S}[1] \rightarrow \dfrac{2\rho}{1+\rho^2}\left(\dfrac{X[0]+X[2]}{2}\right)$ and $\sigma_\varepsilon^2 \rightarrow \sigma_S^2\dfrac{1-\rho^2}{1+\rho^2}$ which are the LMMSE of $S[1]$ and its MSE in the absence of noise.